

CODE SPORT



In this month's column, we discuss a few computer science interview questions.

Over the past couple of months, we have been discussing the problem of detecting duplicate questions in community question answering (CQA) forums using the Quora Question Pair Dataset. In this month's column, we take a break from this topic and, as requested by our readers, discuss a few interview questions, focusing on algorithms, machine learning and text analytics.

1. In machine learning, there are three classes of errors – namely, training error, validation error and test error. Can you differentiate between each of these three classes of errors? During training, we focus on reducing the training error. In real world deployment, when the classifier is employed on unseen data, how can reducing training errors help to predict better?
2. There are two classifiers you have trained for the given image classification problem. Classifier A has a training error of 4.2 per cent and Classifier B has a training error of 0.8 per cent. If you are using Classifiers A and B on a held-out test data set, what can you say about the test accuracies of A and B? Would B have a better test accuracy than A? Provide the rationale behind your answer.
3. Can you explain the concept of generalisation in the context of machine learning? What are the measures that can be used to explain the generalisation ability of learning systems?
4. What is meant by the Vapnik–Chervonenkis (VC) dimension in machine learning? Why is this an important classifier? What is the VC dimension of a linear classifier?
5. Auto-encoders are a class of unsupervised deep learning systems that take an input X and produce an output Y , which is as close to X as possible. Auto-encoders can be a deep multi-layer neural network. Since they try to reproduce the input given to them at the output, are they learning an identity function? What's the rationale behind your answer?
6. As you know, while training a deep learning network, the optimisation objective is to minimise

the loss function defined for that classification problem. Given that non-convex optimisation functions can have multiple local minima, do first-order gradient descent based optimisation methods ensure that we reach a global minimum? If you think they do, explain how. If not, explain why.

7. Distributional semantics is based on the idea that semantic similarities can be characterised based on the distributional properties of the words in a large corpus. This is popularly expressed by the statement, "A word is known by the company it keeps." If two words have similar distributions or similar contexts, then they have similar meanings. Can you give a couple of examples of the applications of distributional semantics in natural language processing?
8. Traditionally, text inputs to statistical NLP systems have used one hot vector representation of words. For instance, given a vocabulary of size V , each word is represented by a vector of size V , where only one index is set to one, corresponding to that word, and all the remaining indices in the vector are set to zero. This is known as the 'one hot vector representation'. However, of late, vector space word models are becoming quite popular as alternative input representations. Models such as Word2Vec (<https://radimrehurek.com/gensim/models/word2vec.html>) and Glove (<http://nlp.stanford.edu/projects/glove/>) are quite popular these days for input representation in statistical NLP systems, as opposed to representing words as one hot vector. What is the advantage of using vector space word models over traditional one hot vector representation? Are there any disadvantages with such vector space models?
9. You are given a corpus and you have created the word embeddings for your corpus using the popular Word2Vec technique. Now you need to represent each sentence in your corpus by means of a vector. You need to compose the word vectors of the words in your sentence to generate a sentence vector